Artificial Intelligence
UNIT IV
Knowledge representation using Logical Formalisms, Propositional and First order Predicate Calculus

Prepared By

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Introduction to Logical Formalisms: Role of logic in Artificial Intelligence, Knowledge-based agents Syntax and semantics of logical systems

Propositional Logic – Basics and Inference: Propositional symbols and well-formed formulas, Logical connectives, Inference rules: Modus Ponens, Modus Tollens, Resolution

First Order Predicate Logic – Fundamentals: Motivation for First Order Logic, Quantifiers (∀, ∃), Well-formed formulas, Translating natural language into FOL

Inference in First Order Predicate Logic – Fundamentals: Motivation for First Order Logic, Quantifiers (∀, ∃), Well-formed formulas, Translating natural language into FOL

Case Study: Medical Expert System – MYCIN, Knowledge-Based Reasoning in Intelligent Search
Systems:- AI-Based Rule Engine in Google Search

Logical Agents

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Role of logic in Artificial Intelligence

Logic acts as the foundational framework in artificial intelligence for representing knowledge, making deterministic inferences, and enabling machines to reason through rules

Core Functions of Logic in AI

Knowledge Representation: Logic allows AI systems to structure facts and relationships about the world into a formal language.

Inference and Decision Making: Using rules of inference, machines derive new conclusions from existing data without guessing.

Explainability: Logic-based systems provide clear, step-by-step audit trails for decisions, avoiding the "black box" problem common in pure statistical models.

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Types of Logic Used

Propositional Logic: Deals with basic statements that are either strictly true or false.

First-Order Predicate Logic: Adds variables, functions, and quantifiers (like "for all" or "there exists") to express complex real-world relationships.

Nonmonotonic Logic: Allows an AI to retract previous conclusions when new, contradicting information arrives.

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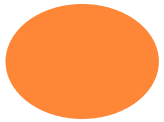
Applications

Expert Systems: Medical or financial software that applies logical rules to diagnose problems or offer advice.

Automated Reasoning Models: Modern AI architectures that use internal step-by-step thinking tokens to solve hard math or logic tasks.

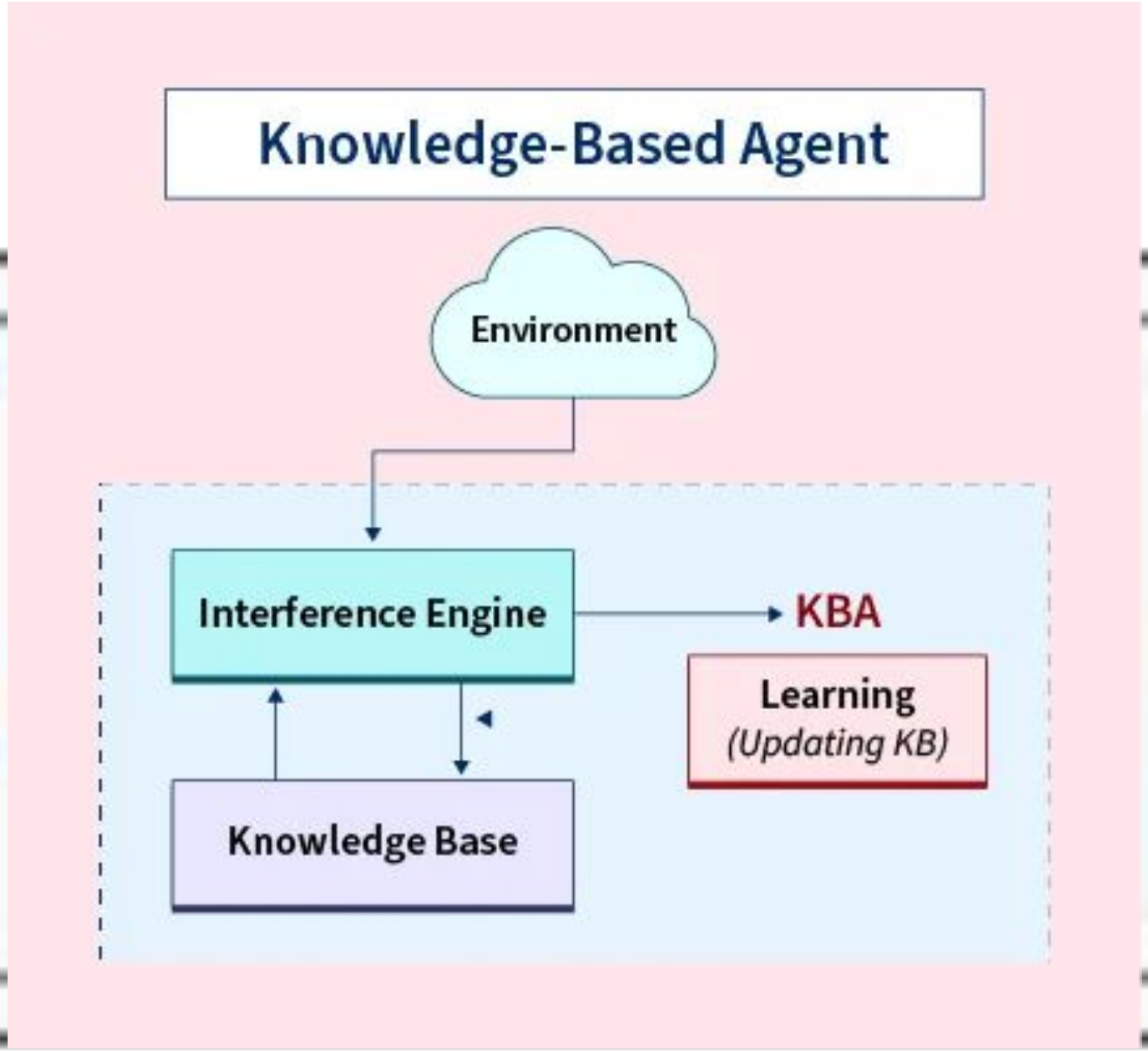
Logic Programming: Languages like Prolog that use defined facts and rules to let systems compute answers natively.

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Knowledge-Based Agents

- Intelligent agents need knowledge about the world in order to reach good decisions.
- In artificial intelligence, a knowledge-based agent uses a central knowledge base of represented facts to reason, make decisions, and learn about its environment.
- Knowledge is contained in agents in the form of **sentences** in a **knowledge representation language** that are stored in a **knowledge base**.
- A knowledge-based agent is composed of a knowledge base and an inference mechanism.
- It operates by storing sentences about the world in its knowledge base, using the inference mechanism to infer new sentences, and using these sentences to decide what action to take.



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Knowledge-Based Agents are Composed of Two Main Parts

Knowledge-Base

The knowledge base of a knowledge-based agent is a collection of knowledge that the agent uses to make decisions. The knowledge can be explicit, such as rules or facts, or implicit, such as relationships or patterns. The knowledge base is stored in a database or a knowledge representation system.

Inference System

The inference system of a knowledge-based agent is responsible for using the knowledge in the knowledge base to make decisions. The inference system uses reasoning methods, such as deduction, induction, and abduction, to infer new knowledge from existing knowledge.

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Logic

- **Logics** are formal languages for representing information such that conclusions can be drawn
- **Syntax** defines the sentences in the language
- **Semantics** define the "meaning" of sentences;
  - i.e., define **truth** of a sentence in a world
- E.g., the language of arithmetic
- $x+2 \geq y$ is a sentence; $x2+y > \{\}$ is not a sentence
- $x+2 \geq y$ is true if the number $x+2$ is not less than the number y
- $x+2 \geq y$ is true in a world where $x = 7, y = 1$
- $x+2 \geq y$ is false in a world where $x = 0, y = 6$
- The relationship of **entailment** between sentences is crucial to our understanding of reasoning.
- A sentence α entails another sentence β if β is true in all worlds where α is true.
- Equivalent definitions include the **validity** of the sentence $\alpha \Rightarrow \beta$ and
- The **unsatisfiability** of the sentence $\alpha \wedge \neg\beta$.

PROPOSITIONAL LOGIC

- **Logical constants**: true, false
- **Propositional symbols**: P, Q, S, ... (**atomic sentences**)
- Wrapping **parentheses**: (…)
- Sentences are combined by **connectives**:

  $\wedge$ ...and [conjunction]

  $\vee$ ...or [disjunction]

  $\Rightarrow$ ...implies [implication / conditional]

  $\Leftrightarrow$ ..is equivalent [biconditional]

  $\neg$ ...not [negation]

- **Literal**: atomic sentence or negated atomic sentence

Examples of PL sentences

- P means “It is hot.”
- Q means “It is humid.”
- R means “It is raining.”
- $(P \wedge Q) \rightarrow R$

  “If it is hot and humid, then it is raining”
- $Q \rightarrow P$

  “If it is humid, then it is hot”
- A better way:

  Hot = “It is hot”

  Humid = “It is humid”

  Raining = “It is raining”

PROPOSITIONAL LOGIC (PL)

- A simple language useful for showing key ideas and definitions
- User defines a set of propositional symbols, like P and Q.
- User defines the **semantics** of each propositional symbol:
  - P means "It is hot"
  - Q means "It is humid"
  - R means "It is raining"
- A sentence (well formed formula) is defined as follows:
  - A symbol is a sentence
  - If S is a sentence, then ¬S is a sentence
  - If S is a sentence, then (S) is a sentence
  - If S and T are sentences, then (S $\vee$ T), (S $\wedge$ T), (S $\rightarrow$ T), and (S $\leftrightarrow$ T) are sentences
  - A sentence results from a finite number of applications of the above rules

A Backus normal form (BNF) grammar of sentences in propositional logic

```
S := <Sentence> ;
<Sentence> := <AtomicSentence> | <ComplexSentence> ;
<AtomicSentence> := "TRUE" | "FALSE" |
                    "P" | "Q" | "S" ;
<ComplexSentence> := "(" <Sentence> ")" |
          <Sentence> <Connective> <Sentence> |
          "NOT" <Sentence> ;
<Connective> := "NOT" | "AND" | "OR" | "IMPLIES" |
  "EQUIVALENT" ;
```

Some terms

- The meaning or **semantics** of a sentence determines its **interpretation**.
- Given the truth values of all symbols in a sentence, it can be "evaluated" to determine its **truth value** (True or False).
- A **model** for a KB is a "possible world" (assignment of truth values to propositional symbols) in which each sentence in the KB is True.

More terms

1. A **valid sentence** or **tautology** is a sentence that is True under all interpretations, no matter what the world is actually like or how the semantics are defined. Example: “It’s raining or it’s not raining.”
2. An **inconsistent sentence** or **contradiction** is a sentence that is False under all interpretations. The world is never like what it describes, as in “It’s raining and it’s not raining.”
3. **P entails Q**, written P |= Q, means that whenever P is True, so is Q must true. In other words, all models of P are also models of Q.

   Let **P** be the statement: **"It is raining heavily and the sky is cloudy."**

   Let **Q** be the statement: **"It is raining."**

   We want to show that **(P |=Q)** (P entails Q). This means in any possible world where P is completely true, Q must also be true.

Truth tables

And

p	q	$p \cdot q$
T	T	T
T	F	F
F	T	F
F	F	F

Or

p	q	$p \vee q$
T	T	T
T	F	T
F	T	T
F	F	F

If . . . then

p	q	$p \rightarrow q$
T	T	T
T	F	F
F	T	T
F	F	T

Not

p	$\sim p$
T	F
F	T

TRUTH TABLES II

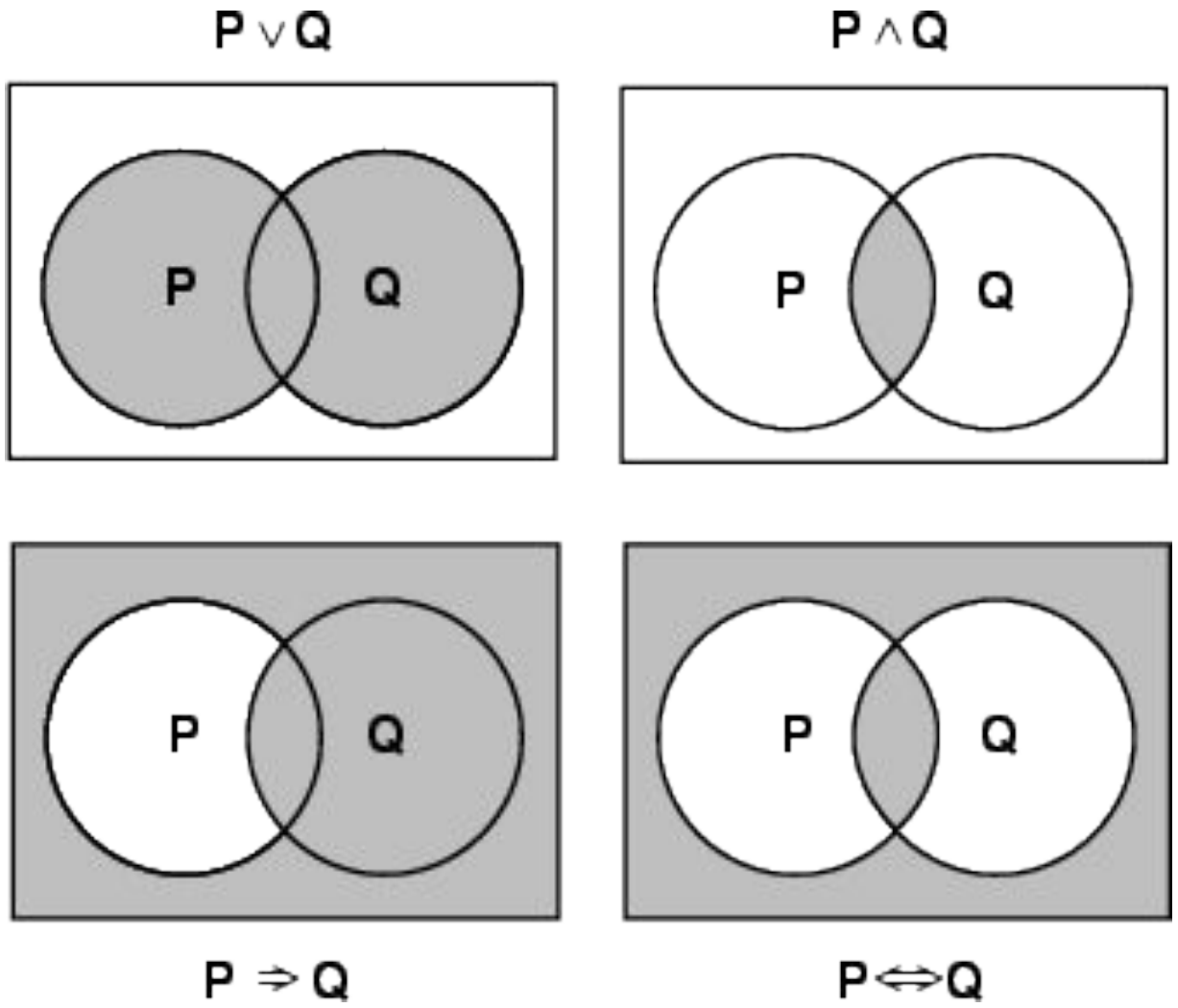
The five logical connectives:

P	Q	$\neg P$	$P \wedge Q$	$P \vee Q$	$P \Rightarrow Q$	$P \Leftrightarrow Q$
False	*False*	*True*	*False*	*False*	*True*	*True*
False	*True*	*True*	*False*	*True*	*True*	*False*
True	*False*	*False*	*False*	*True*	*False*	*False*
True	*True*	*False*	*True*	*True*	*True*	*True*

A complex sentence:

P	H	$P \vee H$	$(P \vee H) \wedge \neg H$	$((P \vee H) \wedge \neg H) \Rightarrow P$
False	*False*	*False*	*False*	*True*
False	*True*	*True*	*False*	*True*
True	*False*	*True*	*True*	*True*
True	*True*	*True*	*False*	*True*

Models of complex sentences



INFERENCE RULES

- Inference rules are the templates for generating valid arguments. Inference rules are applied to derive proofs in artificial intelligence, and the proof is a sequence of the conclusion that leads to the desired goal.

- **Logical inference** is used to create new sentences that logically follow from a given **set of predicate calculus sentences** (KB).

- An inference rule is **sound** if every sentence X produced by an inference rule operating on a KB logically follows from the KB. (That is, the inference rule does not create any contradictions)

- An inference rule is **complete** if it is able to produce every expression that logically follows from (is entailed by) the KB. (Note the analogy to complete search algorithms.)

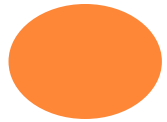
The Scenario

Premise 1: If it is raining, then the ground is wet. ($R \rightarrow W$)

Premise 2: It is raining. (R)

Using these premises, derive the conclusion:

Conclusion: The ground is wet. (W)



In inference rules, the implication among all the connectives plays an important role. Following are some terminologies related to inference rules:

- **Implication:** It is one of the logical connectives which can be represented as P → Q. It is a Boolean expression.
- **Converse:** The converse of implication, which means the right-hand side proposition goes to the left-hand side and vice-versa. It can be written as Q → P.
- **Contrapositive:** The negation of converse is termed as contrapositive, and it can be represented as ¬ Q → ¬ P.
- **Inverse:** The negation of implication is called inverse. It can be represented as ¬ P → ¬ Q.
- From the above term some of the compound statements are equivalent to each other, which we can prove using truth table:

Sound rules of inference

- Here are some examples of sound rules of inference
  - *A rule is sound if its conclusion is true whenever the premise is true*
- Each can be shown to be sound using a truth table

RULE	**PREMISE**	**CONCLUSION**
Modus Ponens	$A, A \rightarrow B$	B
And Introduction	A, B	$A \wedge B$
And Elimination	$A \wedge B$	A
Double Negation	$\neg\neg A$	A
Unit Resolution	$A \vee B, \neg B$	A
Resolution	$A \vee B, \neg B \vee C$	$A \vee C$

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1. Modus Ponens:

- The Modus Ponens rule is one of the most important rules of inference, and it states that if P and P → Q is true, then we can infer that Q will be true. It can be represented as:

Notation for Modus ponens: $\dfrac{P \rightarrow Q, \quad P}{\therefore Q}$

- **Example:**
- Statement-1: "If I am sleepy then I go to bed" ==> P→ Q
  Statement-2: "I am sleepy" ==> P
  Conclusion: "I go to bed." ==> Q.
  Hence, we can say that, if P→ Q is true and P is true then Q will be true.

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Proof by Truth table:

P	Q	$P \rightarrow Q$
0	0	1
0	1	1
1	0	0
1	1	1



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2. Modus Tollens:

- The Modus Tollens rule state that if P→ Q is true and ¬ **Q is true, then ¬ P** will also true. It can be represented as:

Notation for Modus Tollens: $\frac{P \rightarrow Q, \ \sim Q}{\sim P}$

- **Statement-1:** "If I am sleepy then I go to bed" ==> P→ Q
  Statement-2: "I do not go to the bed."==> ~Q
  Statement-3: Which infers that "**I am not sleepy**" => ~P

Proof by Truth table:

P	Q	$\sim P$	$\sim Q$	$P \rightarrow Q$
0	0	1	1	1
0	1	1	0	1
1	0	0	1	0
1	1	0	0	1



3. Hypothetical Syllogism:

- The Hypothetical Syllogism rule state that if P→R is true whenever P→Q is true, and Q→R is true. It can be represented as the following notation:
- **Example:**
- **Statement-1:** If you have my home key then you can unlock my home. **P→Q**
  Statement-2: If you can unlock my home then you can take my money. **Q→R**
  Conclusion: If you have my home key then you can take my money. **P→R**

Proof by Truth table:

P	Q	R	$P \rightarrow Q$	$Q \rightarrow R$	$P \rightarrow R$
0	0	0	1	1	1
0	0	1	1	1	1
0	1	0	1	0	1
0	1	1	1	1	1
1	0	0	0	1	1
1	0	1	0	1	1
1	1	0	1	0	0
1	1	1	1	1	1

4. Disjunctive Syllogism:

- The Disjunctive syllogism rule state that if $P \vee Q$ is true, and $\neg P$ is true, then Q will be true. It can be represented as:

Notation of Disjunctive syllogism: $\dfrac{P \vee Q,\ \neg P}{Q}$

- **Example:**
- **Statement-1:** Today is Sunday or Monday. ==>$P \vee Q$
  Statement-2: Today is not Sunday. ==> $\neg P$
  Conclusion: Today is Monday. ==> Q

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Proof by Truth table:

P	Q	$\neg P$	$P \vee Q$
0	0	1	0
0	1	1	1
1	0	0	1
1	1	0	1



5. Addition:

- The Addition rule is one the common inference rule, and it states that If P is true, then P ∨ Q will be true.

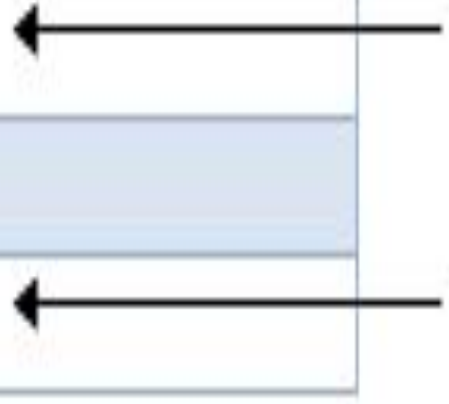
Notation of Addition: $\frac{P}{P \vee Q}$

- **Example:**
- **Statement:** I have a vanilla ice-cream. ==> P
  Statement-2: I have Chocolate ice-cream.
  Conclusion: I have vanilla or chocolate ice-cream. ==> (P ∨ Q)

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Proof by Truth table:

P	Q	P ∨ Q
0	0	0
1	0	1
0	1	1
1	1	1



6. Simplification:

- The simplification rule state that if **P ∧ Q** is true, then **Q or P** will also be true. It can be represented as:

Notation of Simplification rule: $\frac{P \wedge Q}{Q}$ Or $\frac{P \wedge Q}{P}$

P	Q	$P \wedge Q$
0	0	0
1	0	0
0	1	0
1	1	1

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7. Resolution:

- The Resolution rule state that if $P \vee Q$ and $\neg P \wedge R$ is true, then $Q \vee R$ will also be true. **It can be represented as**

Notation of Resolution $\dfrac{P \vee Q,\ \neg P \wedge R}{Q \vee R}$

P	¬ P	Q	R	$P \vee Q$	$\neg P \wedge R$	$Q \vee R$
0	1	0	0	0	0	0
0	1	0	1	0	1	1
0	1	1	0	1	0	1
0	1	1	1	1	1	1
1	0	0	0	1	0	0
1	0	0	1	1	0	1
1	0	1	0	1	0	1
1	0	1	1	1	0	1

Proving things

- A **proof** is a sequence of sentences, where each sentence is either a premise or a sentence derived from earlier sentences in the proof by one of the rules of inference.
- The last sentence is the **theorem** (also called goal or query) that we want to prove.
- Example for the "weather problem" given above.

1 Humid	Premise	"It is humid"
2 Humid→Hot	Premise	"If it is humid, it is hot"
3 Hot	Modus Ponens(1,2)	"It is hot"
4 (Hot∧Humid)→Rain	Premise	"If it's hot & humid, it's raining"
5 Hot∧Humid	And Introduction(1,2)	"It is hot and humid"
6 Rain	Modus Ponens(4,5)	"It is raining"

HORN SENTENCES

Horn clause is clause (a disjunction of literals) with at most one positive, i.e. unnegated, literal. A clause with **at most one positive (unnegated) literal** is called a Horn Clause.

- A **Horn sentence** or **Horn clause** has the form:

  $P1 \land P2 \land P3 \ldots \land Pn \rightarrow Q$

or alternatively

$$(P \rightarrow Q) = (\neg P \lor Q)$$

$\neg P1 \lor \neg P2 \lor \neg P3 \ldots \lor \neg Pn \lor Q$

Where It has only one positive literal Q.

- To get a proof for Horn sentences, apply Modus Ponens repeatedly until nothing can be done

Entailment and derivation

- **Entailment: KB |= Q**
  - Q is entailed by KB (a set of premises or assumptions) if and only if there is no logically possible world in which Q is false while all the premises in KB are true.
  - Or, stated positively, Q is entailed by KB if and only if the conclusion is true in every logically possible world in which all the premises in KB are true.

Knowledge Base (KB):

1. All humans are mortal.
2. Socrates is a human.

Query (Q):

1. Socrates is mortal.

We can write:

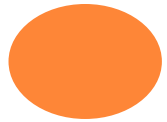
KB ⊨ Q

Because:

1. If **all humans are mortal** is true, and
2. **Socrates is a human** is true,
3. then **Socrates must be mortal**.

There is **no logically possible world** where both KB statements are true but Socrates is not mortal.

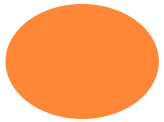
Therefore, **Q is entailed by KB.**



- **Derivation: KB |- Q**
  - We can derive Q from KB if there is a proof consisting of a sequence of valid inference steps starting from the premises in KB and resulting in Q

Derivation = Starting with known facts and using rules of inference to prove the required conclusion.

The symbol ⊢ is read as **“derives”** or **“can be proved from.”**



Example: Modus Ponens

Suppose our Knowledge Base contains:

1. **If it rains, then the ground is wet.**
2. **It is raining.**

Let:

- **P = It is raining**
- **Q = Ground is wet**

So KB contains:

$$P \rightarrow Q \text{ , } P$$

We want to derive:

Q

Derivation steps

Step 1: Start with the first premise:

$P \rightarrow Q$

Step 2: Take the second premise:

P

Step 3: Apply the **Modus Ponens** inference rule:

$$P \rightarrow Q, P$$
$$Q$$

Therefore:

$KB \vdash Q$

That means **Q can be derived/proved from KB**.

Two important properties for inference

Soundness: If KB |- Q then KB |= Q

- If Q is derived from a set of sentences KB using a given set of rules of inference, then Q is entailed by KB.
- Hence, inference produces only real entailments, or any sentence that follows deductively from the premises is valid.

Completeness: If KB |= Q then KB |- Q

- If Q is entailed by a set of sentences KB, then Q can be derived from KB using the rules of inference.
- Hence, inference produces all entailments, or all valid sentences can be proved from the premises.

PROPOSITIONAL LOGIC IS A WEAK LANGUAGE

1. Hard to identify "individuals" (e.g., Mary, 3)

2. Can't directly talk about properties of individuals or relations between individuals (e.g., "Bill is tall")

3. Generalizations, patterns, regularities can't easily be represented (e.g., "all triangles have 3 sides")

4. First-Order Logic (abbreviated FOL or FOPC) is expressive enough to concisely represent this kind of information

   FOL adds relations, variables, and quantifiers, e.g.,

   - *"Every elephant is gray"*: $\forall$ x (elephant(x) $\rightarrow$ gray(x))
   - *"There is a white alligator"*: $\exists$ x (alligator(X) ^ white(X))

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Example

- Consider the problem of representing the following information:
  - Every person is mortal.
  - Newton is a person.
  - Newton is mortal.
- How can these sentences be represented so that we can infer the third sentence from the first two?

1. Represent the statements

Sentence 1: Every person is mortal.

This means **all persons are mortal**.

We use the universal quantifier ∀:

∀x (Person(x)→Mortal(x))

Meaning:

For every person x, if x is a person, then x is mortal.

Sentence 2: Newton is a person.

Person(Newton) This is a **fact** about Newton.

Sentence 3: Newton is mortal.

Mortal(Newton) This is the conclusion we want to derive.

2. Apply inference

Our Knowledge Base is:

KB={∀x(Person(x)→Mortal(x)), Person(Newton)}

Using **Universal Instantiation**, substitute **Newton** for x:

Person(Newton)→Mortal(Newton)

Now we have:

Person(Newton) → Mortal(Newton) , Person(Newton)

Mortal(Newton)

Using **Modus Ponens**:

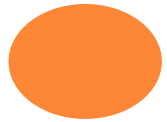
Mortal(Newton)

Therefore:

KB⊢Mortal(Newton)

And because the inference is sound:

KB⊨Mortal(Newton)



Summary

- The process of deriving new sentences from old one is called **inference**.
  - **Sound** inference processes derives true conclusions given true premises
  - **Complete** inference processes derive all true conclusions from a set of premises
- A **valid sentence** is true in all worlds under all interpretations
- If an implication sentence can be shown to be valid, then—given its premise—its consequent can be derived
- Different logics make different **commitments** about what the world is made of and what kind of beliefs we can have regarding the facts
  - Logics are useful for the commitments they do not make because lack of commitment gives the knowledge base engineer more freedom
- **Propositional logic** commits only to the existence of facts that may or may not be the case in the world being represented
  - It has a simple syntax and simple semantics. It suffices to illustrate the process of inference
  - Propositional logic quickly becomes impractical, even for very small worlds